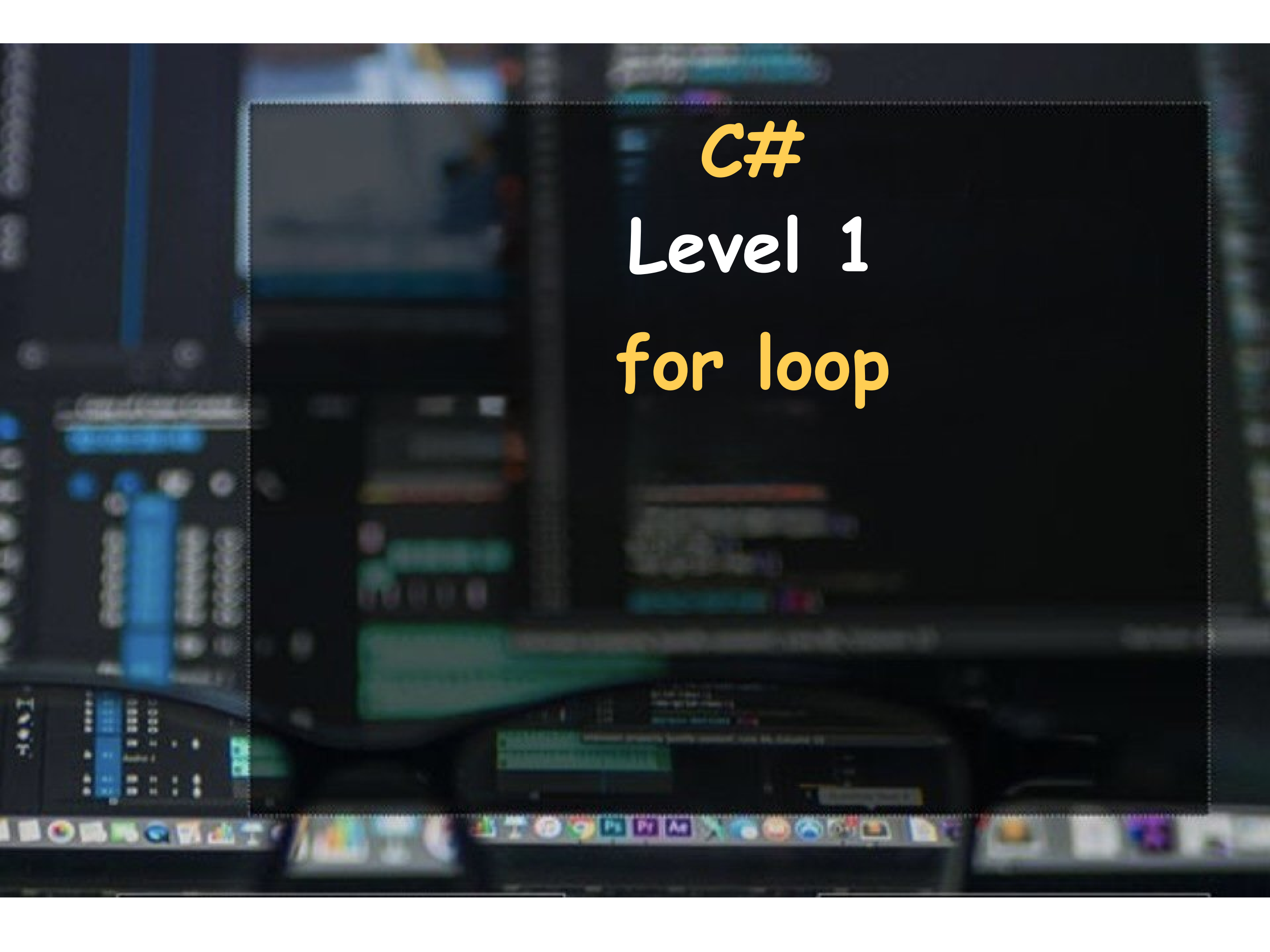
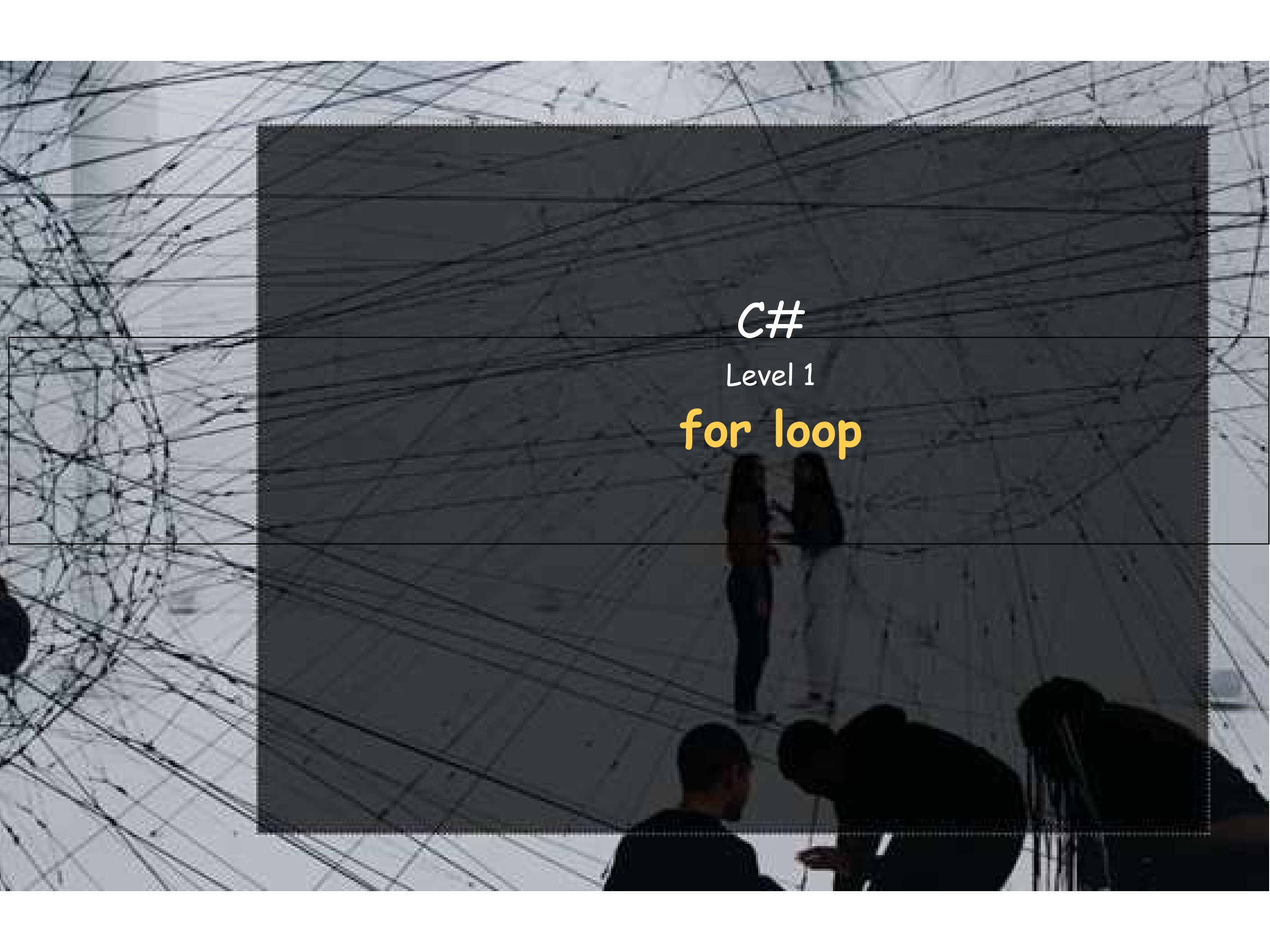




C# Level 1 for loop



C#

Level 1

for loop

Definition

for Loop

-  Repeats a block of code while a condition is true.

-  Commonly used when the number of iterations is controlled by a counter.

Explanation

Three Parts

-  Initialization → runs once before the loop starts.
-  Condition → checked before each iteration.
-  Iterator → runs after each iteration to update the counter.

Forward Loop

-  Starts from a smaller value and increases the counter.
-  Usually uses `i++`.

Backward Loop

-  Starts from a larger value and decreases the counter.
-  Usually uses `i--`.

Nested for Loop

-  A for loop placed inside another for loop.
-  The inner loop completes all its iterations for every iteration of the outer loop.

💡 Example

```
for (int i = 1; i <= 10; i++)  
{  
    Console.WriteLine(i);  
}
```

➡ Result

1

2

3

....

10

💡 Example

```
for (int i = 10; i >= 1; i--)  
{  
    Console.WriteLine(i);  
}
```

➡ Result

10

....

3

2

1

💡 Example

```
for (int i = 1; i <= 10; i++)  
{  
    for (int j = 0; j < 10; j++)  
    {  
        Console.WriteLine("i={0} and j={1}", i, j);  
    }  
}
```

➡ Total Executions
 $10 \times 10 = 100$

Comparison

- ▣ The basic for loop syntax is essentially the same in *C#* and *C++*.
- ▣ The main difference in this example is the output method.

✦ Summary

- ✓ for repeats code based on a condition.
- ✓ Initialization runs once.
- ✓ Condition controls whether the loop continues.
- ✓ Iterator updates the counter after each iteration.
- ✓ `i++` increases the counter, while `i--` decreases it.
- ✓ Nested loops contain one loop inside another.

Thank you 😊